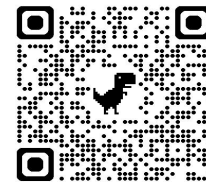


Yuqing Yang

GIS Analyst | Spatial Data Management | Cartography & Web Mapping

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EDUCATION

Master of Urban Spatial Analytics (MUSA)

University of Pennsylvania, GPA: 3.9/4.0

Aug 2025 - May 2026

Philadelphia, PA

Bachelor of Engineering in Urban and Rural Planning

Shenzhen University (SZU)

Sept 2020 - Jun 2025

Shenzhen, China

TECHNICAL SKILLS

GIS & Cartography: ArcGIS Pro, QGIS, coordinate systems and projections, geoprocessing, spatial joins, cartographic design

Spatial Data & Databases: PostgreSQL/PostGIS, SQL, BigQuery, GeoPandas, data cleaning, spatial data integration

Raster & Remote Sensing: GDAL, Rasterio, Google Earth Engine, Sentinel-2, raster processing

Programming & Web Mapping: Python, R, JavaScript, Leaflet, Mapbox API, HTML/CSS, Git/GitHub

PROJECTS

Understanding Off-Peak Roadway Safety (MUSA Practicum)

Jan 2026 - May 2026

University of Pennsylvania | In collaboration with The City of Philadelphia's Office of Transportation and Infrastructure Systems' Office of Multimodal Planning (OTIS)

- Contributed to an end-to-end GIS workflow integrating roadway, speed, crash, road-design, transit, and Google Earth data into a standardized street-segment dataset with 9,049 segment-period observations for citywide analysis.
- Translated analytical results and 24 roadway-intervention scenarios across four daily periods into corridor maps and an interactive decision-support tool, enabling OTIS planners to compare speeding risk and explore potential road-diet, bike-lane, and traffic-calming strategies.

Philadelphia CAMA Pipeline

April 2026 - May 2026

- Contributed to a cloud-native appraisal prototype integrating three citywide datasets—OPA property records, assessment histories, and PWD parcels—through standardized property identifiers and BigQuery source/core/derived schemas.
- Cleaned attributes, connected assessment tables to parcel geometries, and prepared map-ready records for an interactive assessment-review dashboard within a pipeline designed for repeatable weekly data refreshes.

Evaluating Generative Models for Cloud Removal

April 2026 - May 2026

- Developed a reproducible Python/PyTorch benchmark using 128×128 four-band Sentinel-2 image patches, generating controlled cloud masks at 5%, 10%, 30%, 50%, and 70% coverage to compare multi-temporal fusion, GAN-based reconstruction, and mask-guided diffusion inpainting.
- Evaluated 3,000 held-out test cases using PSNR, SSIM, overall and cloud-region L1 error, and NDVI MAE; found that temporal fusion performed strongly under light cloud cover but degraded sharply above 30%, while the GAN maintained substantially more stable reconstruction quality under moderate-to-heavy cloud conditions.

Yunnan Odyssey: Route-Based Travel Planning Web Application

Aug 2025 - Dec 2025

- Built a province-scale, three-stage prototype with a story map, exploration dashboard, and itinerary planner, linking OSM-derived POIs to seasonal filters, search, location-based social content, cost/elevation indicators, and user wishlists.
- Implemented Leaflet, Firebase, Chart.js, and Mapbox routing with day-by-day route generation and drag-and-drop itinerary editing, allowing prospective travelers to move from travel inspiration to mapped plans in one application.

INTERNSHIP

Assistant Urban Analyst, Urban Planning and Natural Resources Bureau of Shenzhen Municipality

Oct 2024 - Jan 2025

- Reviewed planning regulations, converted complex approval requirements into standardized, system-readable components and reusable documentation structures.
- Organized and quality-checked research records, case files, and working documents; prepared templates and process guidance to improve consistency, traceability, and handoff across the digital planning-review workflow.

Urban Planning Intern, Urban Planning & Design Institute of Shenzhen

Jul 2022 - Sep 2022

- Supported vertical and site-planning analysis for Shenzhen airport-area development project and organized technical maps and planning exhibits.
- Compiled and reviewed baseline spatial data for a municipal green-space plan in Zhanjiang City, including verification of planned but unbuilt parks.

AWARDS & HONORS

Published a paper collected by "Proceedings of the 2024 Annual National Planning Conference"

Sept 2023 - Jun 2024

Topic: Migration Characteristics: Comparing Urban Villages and Commercial Housing Residents

Nomination Award at 2024 WUPENiCity International Competition On Urban Sustainability Reports

Shenzhen University | In collaboration with Meituan Academy of Robotics

Sept 2023 - Aug 2024

Topic: A Study On The Suitability Of The Location Of Uavs' Take-Off And Landing Points In The Field Of Real-Time Delivery